Chapter Outline: Valuing Bonds

1. Bond Cash Flows, Prices, and Yields
2. Dynamic Behavior of Bond Prices
3. The Yield Curve and Bond Arbitrage

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Learning Objectives

1. Identify the cash flows for both coupon bonds and zero-coupon bonds, and calculate the value for each type of bond.
2. Calculate the yield to maturity for both coupon and zero-coupon bonds, and interpret its meaning for each.

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Learning Objectives

3. Given coupon rate and yield to maturity, determine whether a coupon bond will sell at a premium or a discount; describe the time path the bond's price will follow as it approaches maturity, assuming prevailing interest rates remain the same over the life of the bond.

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Learning Objectives

4. Illustrate the change in bond price that will occur as a result of changes in interest rates; differentiate between the effect of such a change on long-term versus short-term bonds.
5. Discuss the effect of coupon rate to the sensitivity of a bond price to changes in interest rates.

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1. Bond Cash Flows, Prices, and Yields

- Bond Terminology
  - Bond Certificate
    - States the terms of the bond
  - Maturity Date
    - Final repayment date
  - Term
    - The time remaining until the repayment date
  - Coupon
    - Promised interest payments

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1. Bond Cash Flows, Prices, and Yields (cont'd)

- Bond Terminology
  - Face Value
    - Notional amount used to compute the interest payments
  - Coupon Rate
    - Determines the amount of each coupon payment
  - Coupon Payment

$$CPN = \frac{\text{Coupon Rate} \times \text{Face Value}}{\text{Number of Coupon Payments per Year}}$$

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Zero-Coupon Bonds

- Zero-Coupon Bond
  - Does not make coupon payments
  - Always sells at a **discount** (a price lower than face value), so they are also called **pure discount bonds**
  - **Treasury Bills** are U.S. government zero-coupon bonds with a maturity of up to one year.

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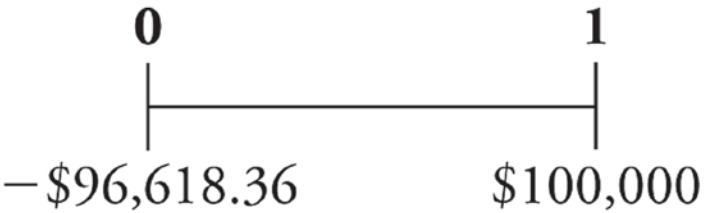
Zero-Coupon Bonds (cont'd)

- Suppose that a one-year, risk-free, zero-coupon bond with a $100,000 face value has an initial price of $96,618.36. The cash flows would be:



  – Although the bond pays no “interest,” your compensation is the difference between the initial price and the face value.

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Zero-Coupon Bonds (cont'd)

- Yield to Maturity
  - *The discount rate that sets the present value of the promised bond payments equal to the current market price of the bond.*
    - *Price of a Zero-Coupon bond*

$$P = \frac{FV}{(1 + YTM_n)^n}$$

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Zero-Coupon Bonds (cont'd)

- Yield to Maturity
  - For the one-year zero coupon bond:

$$96{,}618.36 = \frac{100{,}000}{(1 + YTM_1)}$$

$$1 + YTM_1 = \frac{100{,}000}{96{,}618.36} = 1.035$$

- Thus, the YTM is 3.5%.

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Zero-Coupon Bonds (cont'd)

- Yield to Maturity
  - Yield to Maturity of an *n*-Year Zero-Coupon Bond

$$YTM_n = \left(\frac{FV}{P}\right)^{1/n} - 1$$

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Textbook Example 1

Yields for Different Maturities

Problem

Suppose the following zero-coupon bonds are trading at the prices shown below per $100 face value. Determine the corresponding spot interest rates that determine the zero coupon yield curve.

Maturity	1 Year	2 Years	3 Years	4 Years
Price	$96.62	$92.45	$87.63	$83.06

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Textbook Example 1 (cont'd)

Solution

Using Eq. 6.3, we have

$$r_1 = YTM_1 = (100/96.62) - 1 \quad = 3.50\%$$

$$r_2 = YTM_2 = (100/92.45)^{1/2} - 1 = 4.00\%$$

$$r_3 = YTM_3 = (100/87.63)^{1/3} - 1 = 4.50\%$$

$$r_4 = YTM_4 = (100/83.06)^{1/4} - 1 = 4.75\%$$

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Coupon Bonds

- Coupon Bonds
  - Pay face value at maturity
  - Pay regular coupon interest payments
- Treasury Notes
  - U.S. Treasury coupon security with original maturities of 1–10 years
- Treasury Bonds
  - U.S. Treasury coupon security with original maturities over 10 years

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Textbook Example 2

The Cash Flows of a Coupon Bond

Problem

The U.S. Treasury has just issued a five-year, $1000 bond with a 5% coupon rate and semiannual coupons. What cash flows will you receive if you hold this bond until maturity?

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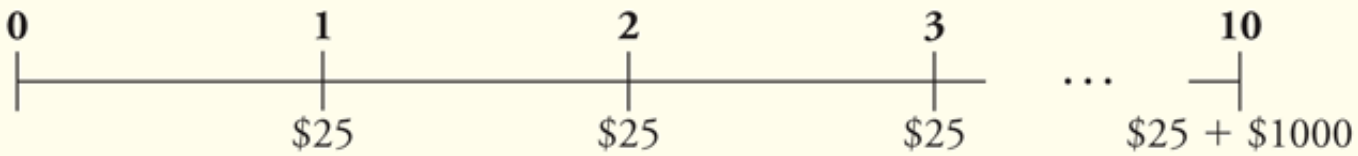
Textbook Example 2 (cont'd)

Solution

The face value of this bond is \$1000. Because this bond pays coupons semiannually, from Eq. 6.1, you will receive a coupon payment every six months of CPN = \$1000 × 5%/2 = \$25. Here is the timeline, based on a six-month period:



Note that the last payment occurs five years (10 six-month periods) from now and is composed of both a coupon payment of \$25 and the face value payment of \$1000.

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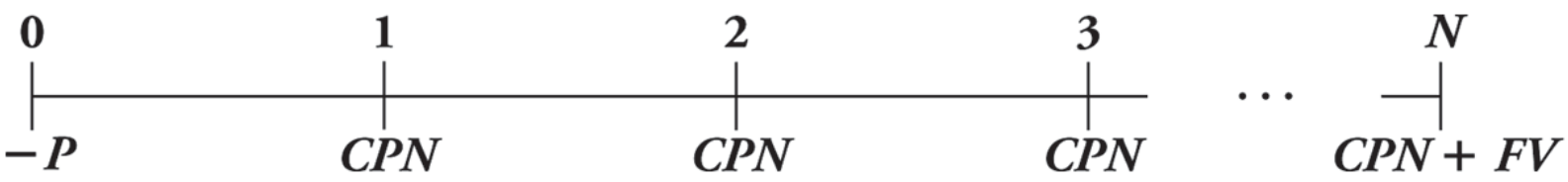
Coupon Bonds (cont'd)

- Yield to Maturity
  - The YTM is the *single* discount rate that equates the present value of the bond's remaining cash flows to its current price.



  - Yield to Maturity of a Coupon Bond

$$P = CPN \times \frac{1}{y}\left(1 - \frac{1}{(1+y)^N}\right) + \frac{FV}{(1+y)^N}$$

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Textbook Example 3

Computing the Yield to Maturity of a Coupon Bond

Problem

Consider the five-year, $1000 bond with a 5% coupon rate and semiannual coupons described in Example 6.2. If this bond is currently trading for a price of $957.35, what is the bond's yield to maturity?

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Textbook Example 3 (cont'd)

Solution

Because the bond has 10 remaining coupon payments, we compute its yield y by solving:

$$957.35 = 25 \times \frac{1}{y}\left(1 - \frac{1}{(1+y)^{10}}\right) + \frac{1000}{(1+y)^{10}}$$

We can solve it by trial-and-error or by using the annuity spreadsheet:

	NPER	RATE	PV	PMT	FV	Excel Formula
Given	10		−957.35	25	1,000	
Solve for PV		**3.00%**				=RATE(10,25,−957.35,1000)

Therefore, $y = 3\%$. Because the bond pays coupons semiannually, this yield is for a six-month period. We convert it to an APR by multiplying by the number of coupon payments per year. Thus the bond has a yield to maturity equal to a 6% APR with semiannual compounding.

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Textbook Example 4

Computing a Bond Price from Its Yield to Maturity

Problem

Consider again the five-year, $1000 bond with a 5% coupon rate and semiannual coupons presented in Example 6.3. Suppose you are told that its yield to maturity has increased to 6.30% (expressed as an APR with semiannual compounding). What price is the bond trading for now?

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Textbook Example 4 (cont'd)

Solution

Given the yield, we can compute the price using Eq. 6.5. First, note that a 6.30% APR is equivalent to a semiannual rate of 3.15%. Therefore, the bond price is

$$P = 25 \times \frac{1}{0.0315}\left(1 - \frac{1}{1.0315^{10}}\right) + \frac{1000}{1.0315^{10}} = \$944.98$$

We can also use the annuity spreadsheet:

	NPER	RATE	PV	PMT	FV	Excel Formula
Given	10	3.15%		25	1,000	
Solve for PV			**−944.98**			=PV(0.0315,10,25,1000)

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2. Dynamic Behavior of Bond Prices

- Discount
  - A bond is selling at a **discount** if the price is less than the face value.
- Par
  - A bond is selling at **par** if the price is equal to the face value.
- Premium
  - A bond is selling at a **premium** if the price is greater than the face value.

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Discounts and Premiums

- If a coupon bond trades at a discount, an investor will earn a return both from receiving the coupons and from receiving a face value that exceeds the price paid for the bond.
  - If a bond trades at a discount, its yield to maturity will exceed its coupon rate.

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Discounts and Premiums (cont'd)

- If a coupon bond trades at a premium it will earn a return from receiving the coupons but this return will be diminished by receiving a face value less than the price paid for the bond.
- Most coupon bonds have a coupon rate so that the bonds will *initially* trade at, or very close to, par.

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Discounts and Premiums (cont'd)

Table 1 Bond Prices Immediately After a Coupon Payment

When the bond price is	We say the bond trades	This occurs when
greater than the face value	"above par" or "at a premium"	Coupon Rate > Yield to Maturity
equal to the face value	"at par"	Coupon Rate = Yield to Maturity
less than the face value	"below par" or "at a discount"	Coupon Rate < Yield to Maturity

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Textbook Example 5

Determining the Discount or Premium of a Coupon Bond

Problem

Consider three 30-year bonds with annual coupon payments. One bond has a 10% coupon rate, one has a 5% coupon rate, and one has a 3% coupon rate. If the yield to maturity of each bond is 5%, what is the price of each bond per $100 face value? Which bond trades at a premium, which trades at a discount, and which trades at par?

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Textbook Example 5 (cont'd)提高coupon v.s. 降低coupon

Solution

We can compute the price of each bond using Eq. 6.5. Therefore, the bond prices are

$$P(10\%\text{ coupon}) = 10 \times \frac{1}{0.05}\left(1 - \frac{1}{1.05^{30}}\right) + \frac{100}{1.05^{30}} = \$176.86 \quad \text{(trades at a premium)}$$

$$P(5\%\text{ coupon}) = 5 \times \frac{1}{0.05}\left(1 - \frac{1}{1.05^{30}}\right) + \frac{100}{1.05^{30}} = \$100.00 \quad \text{(trades at par)}$$

$$P(3\%\text{ coupon}) = 3 \times \frac{1}{0.05}\left(1 - \frac{1}{1.05^{30}}\right) + \frac{100}{1.05^{30}} = \$69.26 \quad \text{(trades at a discount)}$$

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Interest Rate Changes and Bond Prices

- There is an inverse relationship between interest rates and bond prices.
  - As interest rates and bond yields rise, bond prices fall.
  - As interest rates and bond yields fall, bond prices rise.

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Interest Rate Changes and Bond Prices (cont'd)

- The sensitivity of a bond's price to changes in interest rates is measured by the bond's **duration**.
  - Bonds with high durations are highly sensitive to interest rate changes.[到期日愈長, 對利率越敏感]
  - Bonds with low durations are less sensitive to interest rate changes.

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Textbook Example 7

The Interest Rate Sensitivity of Bonds

Problem

Consider a 15-year zero-coupon bond and a 30-year coupon bond with 10% annual coupons. By what percentage will the price of each bond change if its yield to maturity increases from 5% to 6%?

But also need to consider the structure of whole value in bond. For example, value relative high in front and relative high in future.[仍須考量價值在整個債券生命週期的分布]

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Textbook Example 7 (cont'd)

Solution

First, we compute the price of each bond for each yield to maturity:

Yield to Maturity	15-Year, Zero-Coupon Bond	30-Year, 10% Annual Coupon Bond
5%	$\frac{100}{1.05^{15}} = \$48.10$	$10 \times \frac{1}{0.05}\left(1 - \frac{1}{1.05^{30}}\right) + \frac{100}{1.05^{30}} = \$176.86$
6%	$\frac{100}{1.06^{15}} = \$41.73$	$10 \times \frac{1}{0.06}\left(1 - \frac{1}{1.06^{30}}\right) + \frac{100}{1.06^{30}} = \$155.06$

The price of the 15-year zero-coupon bond changes by $(41.73 - 48.10)/48.10 = \boxed{-13.2\%}$ if its yield to maturity increases from 5% to 6%. For the 30-year bond with 10% annual coupons, the price change is $(155.06 - 176.86)/176.86 = \boxed{-12.3\%.}$ Even though the 30-year bond has a longer maturity, because of its high coupon rate, its sensitivity to a change in yield is actually less than that of the 15-year zero coupon bond.

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